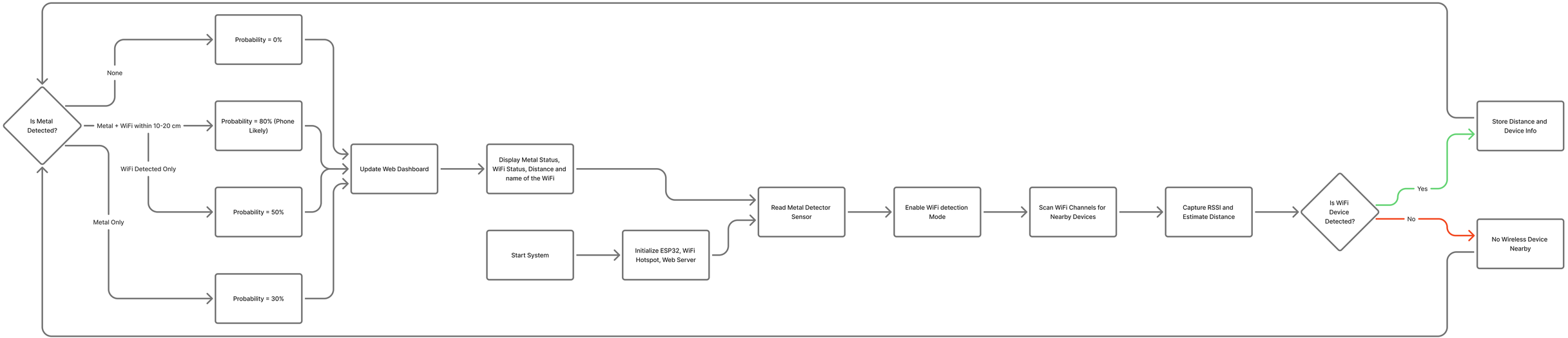
Is Metal Detected?
None
Metal + WiFi within 10-20 cm
WiFi Detected Only
Metal Only
Probability = 0%
Probability = 80% (Phone Likely)
Probability = 50%
Probability = 30%
Update Web Dashboard
Display Metal Status, WiFi Status, Distance and name of the WiFi
Start System
Initialize ESP32, WiFi Hotspot, Web Server
Read Metal Detector Sensor
Enable WiFi detection Mode
Scan WiFi Channels for Nearby Devices
Capture RSSI and Estimate Distance
Is WiFi Device Detected?
Yes
No
Store Distance and Device Info
No Wireless Device Nearby